

PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 4 · Descriptive and probabilistic statistics in sports

TUESDAY SEP 15 AND THURSDAY SEP 17, 2026

WHAT YOU WILL BE ABLE TO DO BY MONDAY NIGHT

- 1** Classify a statistic as counting, efficiency, weighted efficiency or probabilistic, and say what each kind leaves out.
- 2** Name the inputs of a probabilistic statistic, say what it is optimizing for, and say when a team should stop optimizing for it.
- 3** Put two statistics on the same scale with z-scores and explain why they disagree about a play.



UPDATE BOTH FOLDERS BEFORE WE START

1 Open Claude Code in your repo folder, spax402-your-username.

Your spax402 folder holds two folders. This one is your personal repository, where you commit your work. Next to it is spax402-course-materials, our shared materials, which Professor Davis updates weekly.

2 Run one prompt.

The /update skill brings this week's files into your personal repo without saving over your work, and the course materials folder updates by asking Claude to pull the latest version.

Run /update to refresh my repo, then pull the latest course materials in the folder next to mine, spax402-course-materials.



3 Confirm that weeks/week04/ now exists in your repo.

It holds this week's README, the worksheet you fill in on Thursday, and the Case Study data.

01

TWO KINDS OF STATISTIC



WHAT IS THE DIFFERENCE BETWEEN A DESCRIPTIVE STATISTIC AND A PROBABILISTIC ONE?

Points per game. Expected goals. Batting average. Win probability added.

02

COUNTING AND EFFICIENCY

WNBA FINALS, GAME 3, OCTOBER 16, 2024

	MINNESOTA LYNX	NEW YORK LIBERTY
Field goals made	29	28
Field goals attempted	75	61
Three-pointers made	9	10
Free throws made	10	14
Offensive rebounds	7	6
Assists	20	18
Steals	12	8
Turnovers	10	16

Who won the game?



WHAT STAT WOULD YOU USE TO COMPARE TWO BASKETBALL PLAYERS' REBOUNDING ABILITIES?

DESCRIPTIVE STATISTICS

Counting: “Raw” totals accumulated.

Even “per game” numbers are generally considered counting rather than efficiency statistics, as they fail to appropriately adjust for pace, playing time, or usage.

Efficiency: Standardizes counting stats by a common denominator to account for variance in sample size.

Generally an “atomic unit” that is independent of time is preferable.

EFFICIENCY

COUNTING VS STANDARDIZED STATS

COUNTING

Points per game

Strikeouts

Receiving yards

Greens hit in regulation

EFFICIENCY

Points per possession

Strikeouts per 9 innings

Receiving yards per route run

Greens in regulation %

The denominator on the right is the atomic unit: a possession, an inning, a route, a hole.



WHICH OF THE FOUR FACTORS IS MOST IMPORTANT?



WHY ARE WE EXCLUDING ASSISTS, STEALS, AND BLOCKS?



WHAT WOULD YOUR FACTORS BE FOR MEN'S LACROSSE?

03

WEIGHTED EFFICIENCY

TWO HITTERS, 2024

	LUIS ARRAEZ	AARON JUDGE
Team	San Diego Padres	New York Yankees
At bats	637	559
Hits	200 (#2)	180 (#8)
Home runs	4	58 (#1)
Walks	24	133
Batting average	.314 (#4)	.322 (#3)

Batting average says these two had very comparable seasons. Did they?

MLB Stats API, 2024 season hitting totals, qualified hitters. Ranks are among the 129 qualified hitters.

WEIGHTED EFFICIENCY STATISTIC

An efficiency statistic which applies a weight to account for the variance in outcomes.

Batting average weights every hit at 1. Slugging weights a hit by the bases it earns. OPS adds on-base percentage to slugging. wOBA weights every outcome, walks included, by the runs it has historically been worth.

FOUR WEIGHTED EFFICIENCY STATISTICS

$$\text{Slugging \%} = (1B + 2 \times 2B + 3 \times 3B + 4 \times HR) / AB$$

$$OPS = \text{On-base \%} + \text{Slugging \%}$$

$$\begin{aligned} wOBA = & (0.69 \times uBB + 0.72 \times HBP + 0.88 \times 1B + 1.25 \times 2B + 1.58 \times 3B + 2.01 \times HR) \\ & / (AB + BB - IBB + SF + HBP) \end{aligned}$$

$$eFG\% = (2PM + 1.5 \times 3PM) / FGA$$

 **WHICH IS BETTER, OPS OR WOBA?**

04

PROBABILISTIC STATISTICS



WHAT'S THE GOAL OF A PROBABILISTIC STATISTIC?

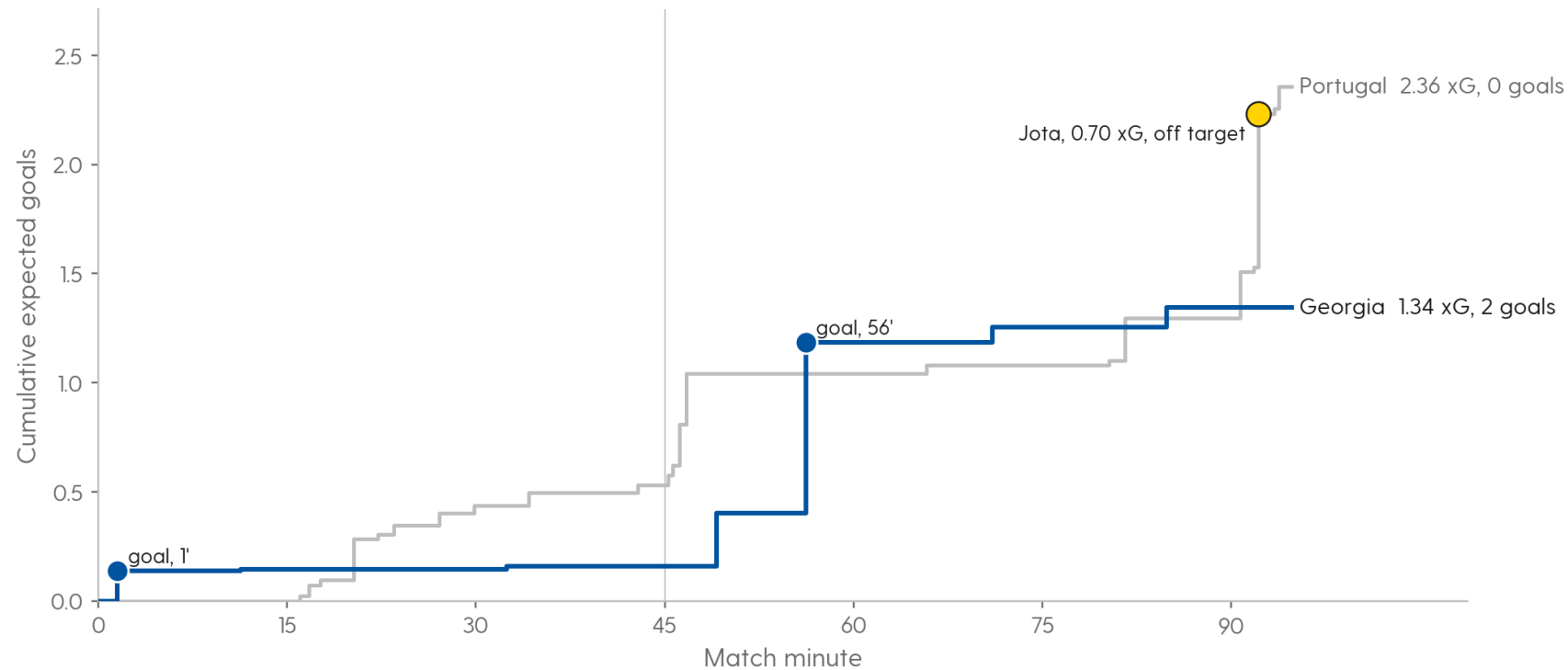
Every descriptive statistic so far counted something that happened. A probabilistic statistic assigns a likelihood to it instead.

EXPECTED GOALS (xG)

The probability that a shot becomes a goal, given where and how it was taken. A team's xG is the sum over its shots.

Inputs are the things about a shot that were true before the ball left the foot: distance and angle to goal, body part, the type of play, the positions of the goalkeeper and defenders. The finish is deliberately left out, because the finish is the thing being measured.

GEORGIA 2, PORTUGAL 0, EURO 2024, JUNE 26, 2024



Portugal put up 2.36 expected goals on 22 shots and scored none. Georgia scored twice on 1.34, and 0.78 of that was one penalty.

StatsBomb open data, Euro 2024, match 3938644. Shot-level xG is StatsBomb's model; the highlighted shot is the highest-xG miss of the match.

NOT ALL PROBABILISTIC STATISTICS ARE CREATED EQUAL. IT'S CRITICAL TO UNDERSTAND THE INPUTS!

One xG model uses shot location alone. Another adds the goalkeeper's position and the defenders between the shooter and the goal. Both are called xG, and they will disagree about the same shot.



WHAT ARE WE OPTIMIZING FOR?

Soccer? Football? Basketball?

WHAT ARE WE OPTIMIZING FOR?

SPORT	OUTCOME
Soccer	Expected goals? Expected goal differential?
Football, basketball	Expected points? Expected point differential?
Baseball, softball	Expected runs? Expected run differential?

GAME CONTEXT MATTERS.

Portugal arrived with 6 points from two games and a knockout place already secured. Georgia needed a win and scored in the second minute. The xG model knows neither fact, and neither team was chasing expected goals after that.

EVEN BASED ON THIS MATCH ALONE, WOULD YOU PICK GEORGIA TO BEAT PORTUGAL IF THEY PLAYED AGAIN?

What would you want to know about the 22 Portugal shots before you answered?

Which inputs to xG would change if the match were played with nothing at stake for either side?

05

EPA AND WPA

TWO WAYS TO BUILD A PROBABILISTIC STATISTIC.

REPLACE THE OUTCOME

Keep what was true before the event, swap the actual result for the average one. Two shots from the same spot get the same number whether one scored and one hit the post.

Expected goals, expected batting average

VALUE THE STATES

Average the eventual points from the state before the play and the state after it, then subtract. Two plays from the same down and distance get different numbers, because the play decides where you land.

Expected points added, win probability added

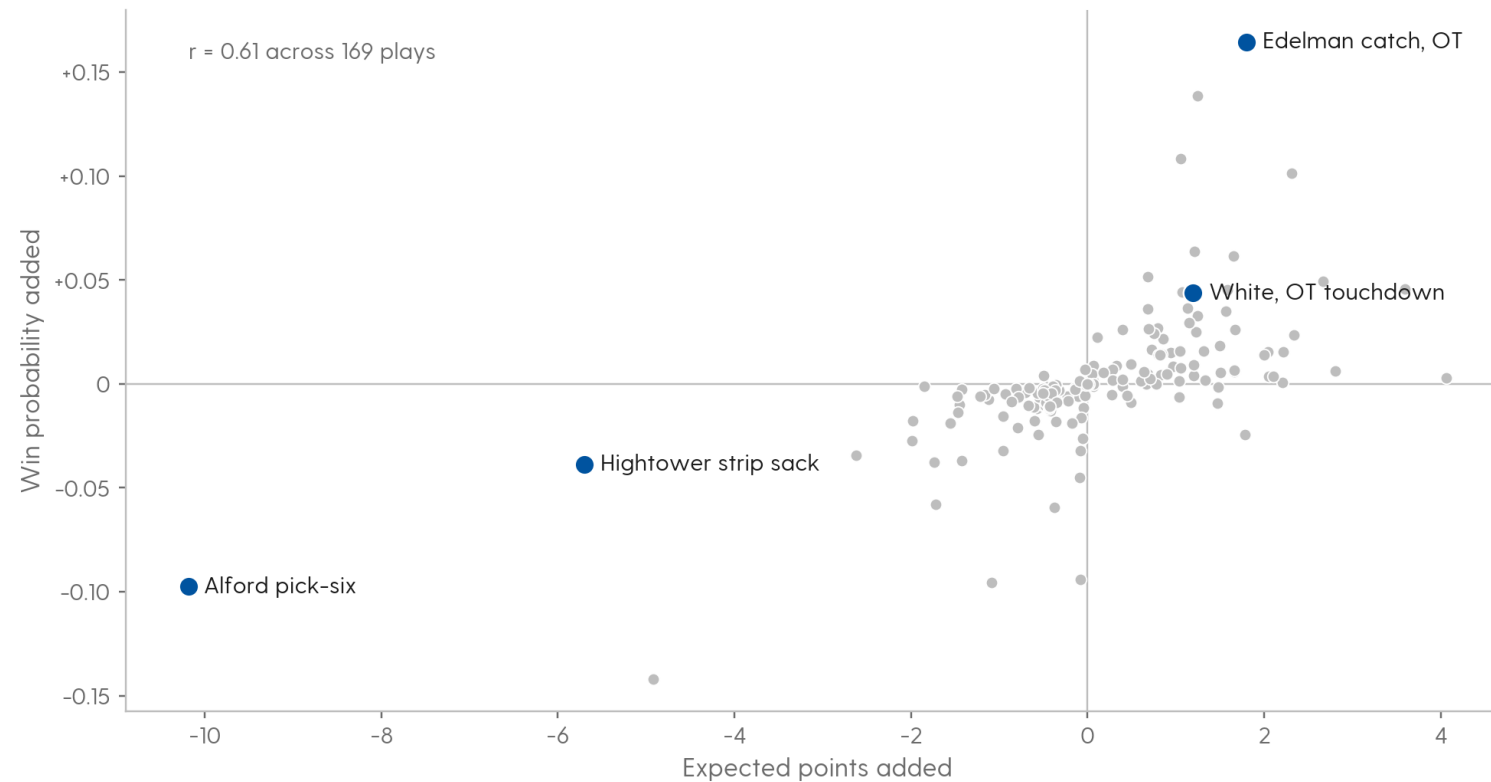
TWO PROBABILISTIC STATISTICS FOR THE SAME PLAY

	EXPECTED POINTS ADDED	WIN PROBABILITY ADDED
Traditional inputs	Possession, yard line, down, distance to go, time remaining	Possession, yard line, down, distance to go, time remaining, score differential
Output	Expected points for the game's next score. Negative weight given to the probability of the opponent scoring next.	Win probability for the game
Added	Expected points after the play minus before	Win probability after the play minus before



**WHAT PLAYS WILL RECEIVE VERY DIFFERENT VALUES ACCORDING TO
EPA AND WPA?**

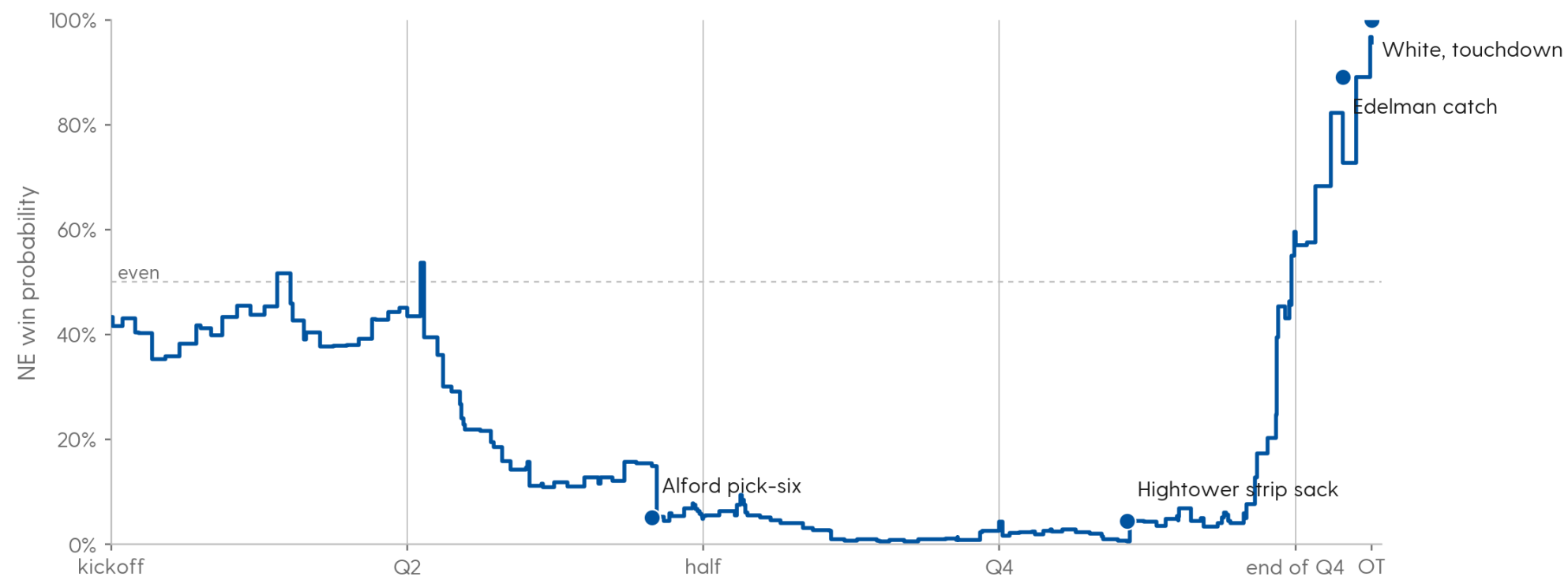
SUPER BOWL LI: NE 34, ATL 28 IN OVERTIME



The two agree on most of 169 plays, $r = 0.61$. The plays that matter are the ones off the line.

nflverse play-by-play, 2016 and 2025 seasons (EPA and WPA are nflfastR's models). EPA and WPA are from the offense's point of view on every play.

NE WIN PROBABILITY, PLAY BY PLAY



NE trailed by 25 points in the third quarter. EPA does not account for the score when valuing the Alford pick-six.

nflverse play-by-play, 2016 and 2025 seasons (EPA and WPA are nflfastR's models).



ALFORD'S PICK-SIX LATE IN THE SECOND QUARTER, OR EDELMAN'S CATCH IN OVERTIME. WHICH WAS THE BIGGER PLAY?

By EPA the interception was -10.19 and the catch was $+1.80$. By WPA the interception was -0.098 and the catch was $+0.165$.

WHERE THE TWO Z-SCORES DISAGREE MOST, SUPER BOWL LI

CLOCK	OFFENSE	SCORE	PLAY	Z EPA	Z WPA	GAP
Q2 02:36	NE	0-14	T.Brady pass intended for D.Amendola INTERCEPTED...	-6.9	-3.0	+3.9
OT 12:37	NE	28-28	T.Brady pass to J.Edelman to ATL 25 for 15 yards	1.2	5.0	+3.9
OT 13:59	NE	28-28	T.Brady pass to C.Hogan to ATL 37 for 18 yards	0.8	4.2	+3.5
Q4 00:18	ATL	28-28	M.Ryan pass incomplete to A.Hooper	-0.1	-2.9	-2.8
Q4 08:31	ATL	28-12	M.Ryan sacked at ATL 25 for -11 yards. FUMBLES...	-3.9	-1.2	+2.7
OT 14:26	NE	28-28	T.Brady pass to D.Amendola ran ob at NE 45 for 14...	0.7	3.3	+2.6

Every large gap is a late play or a scoring play. WPA asks when; EPA asks what.

Gap is z(WPA) minus z(EPA). nflverse play-by-play, 2016 and 2025 seasons (EPA and WPA are nflfastR's models).

WHY THEY DISAGREE

Score differential.

Hightower's strip sack at Q4 08:31 was -5.70 EPA on a drive that would likely have ended the game. WPA was only -0.039, because a team trailing by two scores with eight minutes left has very little win probability to lose.

Time remaining, in a game that can end.

Edelman's OT 12:37 catch in overtime was worth +1.80 expected points and +0.165 of a win, because the next score ended the game.

Neither stat is “better”, but it's important to know what they measure and to use the right tool for the job.

EPA grades the play. WPA grades the play in its moment. A coach evaluating a quarterback wants the first; a coach deciding whether to go for it wants the second.

06

THIS WEEK'S WORK



INDIANA FEVER 76, LOS ANGELES SPARKS 75, AUGUST 29, 2025

1 Open the four factors sheet in the worksheet.

Both teams' box score lines are given, Sparks in column B and Fever in column C, with the 0.44 free throw weight in its own cell.

2 Compute eFG%, FTAr, TO% and ORB% for each team.

Then possessions and points per possession.

3 Say which team won each factor.

LAS shot much better from the field and lost by one. Be ready to say which factor made up the difference, and by how much.



HOW TO BUILD THE FOUR FACTORS

Possessions = FGA - ORB + TOV + 0.44 x FTA.

This formula can be used to estimate possessions when you do not have the data.

Points per possession = PTS / possessions.

This is points per game with the pace taken out of it.

Each factor is a ratio of cells that are already on the sheet.

Build the four with cell references. The 0.44 is a given cell, so you can change it and watch the turnover rate move.



THREE HITTERS YOU HAVE NOT SEEN YET, 2024

1 Open the weighted efficiency sheet.

Jonathan India, Ceddanne Rafaela and Shea Langeliers. Each stat line is given: at bats, hits, doubles, triples, home runs, walks, intentional walks, hit by pitches and sacrifice flies. The six wOBA weights are given below them.

2 Compute singles, then batting average, on-base, slugging and OPS for each.

Singles come first, because nothing else on the sheet works without them.

3 Compute wOBA, then rank the three by average, by OPS and by wOBA.

The three rankings do not agree. Two of these hitters had the same season by average and by slugging, so decide whether wOBA's extra work bought anything OPS did not.



HOW TO BUILD OPS AND WOBA

Singles are not on the stat line. Singles = $H - 2B - 3B - HR$.

The first computed cell on the sheet, and every weight below it depends on it.

Slugging is a SUMPRODUCT of the four hit types against the weights 1, 2, 3, 4. OPS adds on-base percentage to it.

Divide slugging by at bats. On-base has a larger denominator, because walks, hit by pitches and sacrifice flies are not at bats.

wOBA is a SUMPRODUCT of six outcomes against six given weights, over a bigger denominator.

The denominator adds walks, sacrifice flies and hit by pitches to at bats, and takes intentional walks back out.



TWELVE PLAYS FROM A GAME WE DID NOT WORK, BAL 40 AT BUF 41

1 Open the epa and wpa sheet.

Twelve plays are given with their EPA, their WPA and the score they were run at. The mean and standard deviation of each column over the twelve are yours to compute.

2 Compute the z-score of each play's EPA and WPA with STANDARDIZE, and the gap.

Then the correlation between the two columns with CORREL.

3 Find the play with the largest gap and say in one sentence why the two models disagree about it.

Nothing on this game is in today's slides. The Case Study asks the same question about a third game and all of its plays.



HOW TO STANDARDIZE AND CORRELATE

STANDARDIZE(x, mean, standard_dev) is the z-score of one value.

Compute the mean and standard deviation of each column once, with AVERAGE and STDEV.S, and point every row at those two cells.

CORREL(range1, range2) is r, the correlation between two columns.

It runs from -1 to 1 and it is the same whether or not you standardized first.

The gap is the difference of the two z-scores, and the interesting rows are the largest in absolute value.

ABS and MAX find the largest one; sorting by the gap column finds the rest.



YOU WORK FOR A HEAD COACH WHO WANTS TO KNOW WHAT DECIDED SUPER BOWL LX

- 1 Across the whole game, how closely do EPA and WPA agree, and what does that tell you about what each one is measuring?
- 2 On the plays where EPA and WPA disagree most, why do the two models see the same play so differently?
- 3 Where in this game do a counting statistic and an efficiency statistic tell two different stories, and which one do you believe?
- 4 With two minutes left in a one-score game, which of EPA or WPA should I be looking at, and why?
- 5 Which single play would you show the team on film, and what would you say about it?
- 6 What is one thing these two statistics cannot tell me about this game?

The data is `weeks/week04/data/super-bowl-lx-plays.csv`: every play of the game with its EPA, WPA and the win probability before it. Your brief is written to the head coach. `python3 scripts/new_brief.py week04` writes `BRIEF.md` with these as headings; `/coach-brief 4` critiques a draft and will not write one.

BEFORE YOU PUSH ON MONDAY, SEPTEMBER 21

Commit the filled-in worksheet in `weeks/week04/data/`.

Commit your analysis code and its outputs in `weeks/week04/outputs/`.

Answer every question in `BRIEF.md`, in your own words.

Run `/audit`, then `/quiz-me`.

Upload your repository's link to Canvas by Monday, September 21 at 11:59pm.